Chapter 5

Equivalence Relations

Assume that you go to buy a new car. You examine the possible choices, and gradually narrow the scope of your search. You are concerned about the particular make and model of a car, its particular assortment of added features and perhaps even its particular color. But among all the cars that are presently for sale, you will consider ones of the same make, model, features and color to be "equivalent" since they are, for you, totally indistinguishable. Auto companies produce thousands of individual cars every year, but only a handful of essentially different kinds. Fortunately you only need to choose among the handful of different kinds, not among the thousands of different individual cars. Once you choose a particular class of "equivalent" cars (for example, blue Subaru Foresters with manual transmission that have no extra added features), you will be happy with whichever member of that "equivalence class" that you can buy at an acceptable price.

In this chapter we will examine what it means for two things to be in some useful sense "equivalent". This notion turns out to have important applications in mathematics and science, and particularly in group theory. There are many occasions when we want to describe certain objects as being "equivalent". For example, two equations in variables x and y can be thought of as equivalent if they have the same sets of solutions. Thus $y = 3x + 4$ and $6x = 2y - 8$ are equivalent. Two computer programs might be thought of as "equivalent" if they always produce the same output from the same input. In a different context, we might think of two compound sentences in propositional logic to be "equivalent" if they have the same truth tables. For example, the truth table below shows that every implication is

p	q	$p \to q$	$\neg q \to \neg p$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

Table 5.1: 1 is True; 0 is False

equivalent to its contrapositive.

The significance of truth table equivalence can be easily seen if we limit our attention to the set of compound sentences with a single variable p. There are still infinitely many different compound sentences that we can write down, for example,

$$((p \vee p) \wedge (\neg p \implies (p \wedge p))) \implies ((\neg p \vee p) \wedge p).$$

But there are only four possible different truth tables, given by the four compound sentences

p, $\neg p$, $p \vee \neg p$ and $p \wedge \neg p$. Thus every compound sentence with one variable p is equivalent

p	p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
0	0	1	1	0
1	1	0	1	0

to one of these four.

All of the above notions of "equivalent" have something in common. In each case there is a set (cars, equations, programs, compound propositional sentences) which is broken up into disjoint bunches so that two members are "equivalent" exactly when they are in the same bunch. In each case the bunches form a partition of the underlying set.

A partition of a set A, which consists of a set of subsets of A, is often a rather awkward thing to talk about. It is usually much more convenient to talk about the associated relation of being "equivalent" among certain pairs of elements of A. In order to discuss this idea, we define a **binary relation** on a set A to be a subset $\equiv$ of $A \times A$. For $x, y \in A$ we often write "$x \equiv y$" to mean that $(x,y) \in \equiv$. For example, the less than or equal to relation $\leq$ on the set $\mathbf{R}$ of real numbers is formally defined as

$$\leq = \{\, (x,y) \mid x \text{ is less than or equal to } y \,\}.$$

We normally prefer to write "$3 \leq 7$" instead of the rather strange looking, but formally correct, "$(3,7) \in \leq$". The equality relation $=$ on A is defined as $a = b$ if a and b represent the same element of A.

Associated with any partition $\mathscr{P}$ of A is a relation $\equiv_{\mathscr{P}}$ of "equivalence" on A: elements x and y of A are "equivalent", that is, $x \equiv_{\mathscr{P}} y$, if x and y are in the same member of $\mathscr{P}$. A relation $\equiv_{\mathscr{P}}$ that arises out of a partition $\mathscr{P}$ in this way is called an **equivalence relation** on A. In this case the member of $\mathscr{P}$ containing an element $x \in A$ is called the **equivalence class** of x.

Consider, for example, the set A of children in the elementary schools. For certain purposes it is helpful to think of two children as being equivalent if they were born in the same year. We then have one grade for each equivalence class. If Charlie is in the fourth grade, his equivalence class is the fourth grade class. Or we might think of two children as equivalent if they are working at the same grade level in mathematics. We could then partition the children into different math classes, each working at a different level. Alternately, we could think of two children as equivalent if they have the same favorite hobby, and then partition them into clubs that each consist of children with a particular hobby. But sometimes we just need to realize that each child is unique, and focus on the partition with a single child in each equivalence class. These are all different partitions, or equivalence relations, on the same set A. There is no right meaning of "equivalence"; rather, each notion of equivalence is useful in a different context.

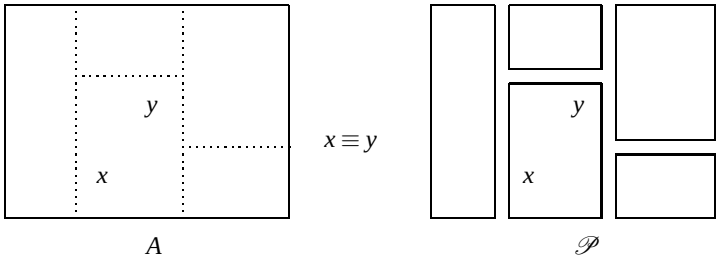
We would like to determine when a binary relation $\equiv$ is the equivalence relation associated with some partition. For example, is the relation $\leq$ on the numbers is an equivalence relation? It is easy to see that every equivalence relation on A must have three simple properties: a relation $\equiv$ on A is

- **reflexive** if $x \equiv x$ for any $x \in A$;
- **symmetric** if $y \equiv x$ whenever $x \equiv y$;
- **transitive** if $x \equiv z$ whenever $x \equiv y$ and $y \equiv z$.

If $\equiv$ is the equivalence relation $\equiv_{\mathscr{P}}$ for some partition $\mathscr{P}$, then it will certainly have these properties. The relation $\leq$, for example, is clearly not symmetric since $4 \leq 9$ but $9 \not\leq 4$. Thus $\leq$ is not an equivalence relation.

Conveniently, it turns out that every binary relation $\equiv$ on A which has these three simple properties is the equivalence relation associated with some partition. To see this, we define—for a binary relation $\equiv$ on A and an element $x \in A$—the **$\equiv$–class** of x to be the set

$$[x] = \{y \mid y \in A \text{ and } x \equiv y\}.$$



Equivalence Relation Theorem *Let $\equiv$ be a binary relation on the set A. If $\equiv$ is reflexive, symmetric and transitive, then the set $\mathscr{P}$ of $\equiv$–classes is a partition of A and $\equiv$ is the equivalence relation $\equiv_{\mathscr{P}}$ associated with $\mathscr{P}$.*

In the first four problems, verify that the relation is an equivalence relation and then describe its equivalence classes, which should form a partition of A.

Problem 50. *A is the set of points (x,y) in the plane. $(x,y) \equiv (x',y')$ if (x,y) and (x',y') are the same distance from the origin.*

Problem 51. *$x \equiv y$ if x and y are integers and $x-y$ is a multiple of 3.*

Problem 52. *$x \equiv y$ if x and y are integers and $x+y$ is even.*

Problem 53. *Let $\mathbf{S} = \{a,b,c,d,e,f,g,h\}$, let $\mathbf{T} = \{a,b,c\}$ and let A be the set of all (2^8) subsets of $\mathbf{S}$. For $X,Y \in A$, we define $X \equiv Y$ if X and Y have the same intersection with $\mathbf{T}$.*

Problem 54. *Let $A = \{0,1\}^5$ be the set of 32 different five-tuples of 0s and 1s, that is, all sequences (a,b,c,d,e) where $a,b,c,d,e \in \{0,1\}$. For $x,y \in A$, define $x \equiv y$ to mean that x and y have the same number of 1s. Show that $\equiv$ is an equivalence relation and write down all the members of each of the different $\equiv$–classes.*

Every function $f : A \to B$ induces a natural equivalence relation on its domain. For $x,y \in A$, we say that x is f–equivalent to y if $f(x) = f(y)$. The classes of this equivalence relation are called **f-classes** of A. For example, the equivalence classes of Problem 50 are exactly the f-classes where $f(x,y) := \sqrt{x^2+y^2}$.

Problem 55. *For each equivalence relation $\equiv$ above, define a function f on A so that the $\equiv$-classes are exactly the f-classes.*

Let $\mathbf{G}$ be a group and let $\mathbf{H}$ be a subgroup of $\mathbf{G}$. We have seen that the right (and left) cosets of $\mathbf{H}$ (both) partition $\mathbf{G}$. These partitions can be described as equivalence relations that $\mathbf{H}$ induces on $\mathbf{G}$. For $a,b \in \mathbf{G}$, we define

$$a \sim_H b \text{ iff } ab^{-1} \in \mathbf{H} \quad \text{and} \quad a_H\!\sim b \text{ iff } a^{-1}b \in \mathbf{H}.$$

Theorem 56. *If* **H** *is a subgroup of* **G**, *then*

(i) $\sim_H$ *is an equivalence relation on* **G**, *and the* $\sim_H$*–classes are exactly the right cosets of* **H**, *i.e.,* $a \sim_H b \Leftrightarrow \mathbf{H}a = \mathbf{H}b$;

(ii) ${}_H\sim$ *is an equivalence relation on* **G**, *and the* ${}_H\sim$*–classes are exactly the left cosets of* **H**, *i.e.,* $a {}_H\sim b \Leftrightarrow a\mathbf{H} = b\mathbf{H}$.

There is a standard and important notion of equivalence between sets. Let A denote the collection of all sets. For $X, Y \in A$, we say that X and Y are the **same size** if there is a bijection (one-to-one and onto function) $f : X \to Y$ from X to Y. Thus finite sets X and Y are the same size if and only if they have the same number of elements, but in general they need not be finite.

Lemma 57. *Let* X, Y *and* Z *be sets.*

(i) *The identity function* $i_X : X \to X$, *with* $i_X(x) = x$, *is a bijection.*

(ii) *If* $f : X \to Y$ *is a bijection, then* $f^{-1} : Y \to X$ *is a bijection.*

(iii) *If* $f : X \to Y$ *and* $g : Y \to Z$ *are bijections, then* $g \circ f : X \to Z$ *is a bijection.*

Theorem 58. *If* X *is a set, the set* $\mathbf{S}_X$ *of all bijections from* X *to itself (called the* **permutations** *of* X*) is a group under composition.*

Theorem 59. *Being the same size is an equivalence relation on the class of all sets.*

What are the equivalence classes of sets in Theorem 59?